Project: AmritaMaa

Smart Dairy Ecosystem — Milk Distribution & Marketplace Platform

(Connecting Farmers, Collection Centers, Delivery Partners & Customers)

Problem Statement

AmritaMaa is a smart digital platform connecting farmers, milk collection centers, delivery partners and customers

- ensuring transparent payments, milk quality monitoring, efficient delivery, and a trusted farm-to-family dairy supply chain.

The current dairy supply chain suffers from several critical gaps that AmritaMaa is designed to solve:

- **Middlemen Dependency —** heavy reliance on middlemen reduces what farmers actually earn.
- **Limited Product Info —** customers have no way to verify the source or quality of the milk they buy.
- **Inconsistent Quality Checks —** the absence of consistent quality monitoring erodes trust across the chain.
- **Poor Transport & Storage —** weak logistics and cold-chain gaps lead to spoilage and loss.
- **No Unified Platform —** no single system links the farmer to the end customer.

Underlying Factors

- Farmers earn less due to continued dependency on middlemen for selling their produce.
- Customers have no reliable way to verify the source or quality of the milk they purchase.
- Lack of consistent quality monitoring builds distrust between farmers, collection centers and customers.
- Weak transport and storage infrastructure leads to spoilage and wastage of milk.
- Farmers, collection centers, delivery partners and customers currently operate in silos with no unified digital system linking them together.

Stakeholders

Internal Users:

- Admin
- Milk Collection Center Staff
- Delivery Partners
- Dairy Company Staff
- Customer Support Staff

External Users:

- Dairy Farmers
- Customers
- Dairy Cooperatives
- Government Departments (Veterinary & Animal Healthcare)
- Dairy Processing Companies

Core Stakeholders: Dairy Farmer • Milk Collection Centers • Delivery Partners • Customer • Admin

Objectives

- **Quality Monitoring —** monitor and maintain milk quality from farm to customer.
- **Reduce Wastage —** reduce milk wastage through better collection and delivery management.
- **Digital Record Keeping —** maintain digital records of milk collection, quality and payment.
- **Traceable Purchases —** enable customers to purchase fresh, safe and traceable milk.

- **Unified Coordination —** enable real-time coordination between farmers, delivery partners, admin and cooperatives.

Proposed Solution & Expected Outputs

AmritaMaa is a unified digital ecosystem that digitizes milk collection, quality grading, delivery logistics and sales — giving every stakeholder a real-time, transparent view of the supply chain.

- **Farmer App —** log daily milk collection, view quality grade & receive instant payments.
- **Collection Center Dashboard —** record Fat% / SNF quality tests and manage daily entries.
- **Delivery Partner App —** optimized delivery routes with live pickup and drop tracking.
- **Customer Marketplace App —** subscribe, order and trace milk from farm to doorstep.
- **Also included —** an Admin Control Panel plus a Government / Cooperative Portal for veterinary and compliance data.

Strategic Information

- State-wise production analysis.
- Farmer income growth.
- Customer satisfaction trends.

Tactical Information

Data flows from farmer collection through quality grading into a central cloud database, where delivery routes are optimized before dispatch to the customer app.

- Daily collection report.
- Quality report (Fat%, SNF).
- Farmer payment status.

Operational Information

- **Real-time Sync —** live inventory & order sync across all apps.
- **Offline-first Design —** the collection app works in low-connectivity areas.
- **Role-based Access —** secure, permissioned dashboards for each stakeholder.
- Day-to-day logs: customer complaints, daily collection entries, and quality test entries.

Feasibility Study

Technical Feasibility:

- Frontend Web — HTML5, Tailwind CSS, React.js.
- Mobile App — React Native, Flutter.
- Backend — Node.js, Express.js, REST APIs.
- Database — MongoDB, MySQL.
- Cloud & DevOps — AWS, Firebase, Git & GitHub.
- Tools & Design — Figma, Postman, VS Code.

- Familiar tech stack (React, Node.js) speeds up development.

Economic Feasibility:

- Direct farmer-to-consumer channel increases farmer income.
- UPI integration leverages India's mature digital payment rails.
- Scales via existing dairy cooperative networks, keeping rollout cost low.

Operational Feasibility:

- Digital quality records (Fat%, SNF) build consumer trust.
- Farmer digital literacy addressed through a simple UI and local training.
- Rural network gaps addressed through an offline-first collection app.
- Trust & adoption built by piloting with government cooperatives first.
- Data privacy ensured through role-based access & encrypted storage.
- Cold-chain cost managed through a phased rollout at high-volume centers.

Requirements

Functional Requirements:

- Farmer milk collection logging & quality grading.
- Instant, UPI-based farmer payments.
- Collection center Fat% / SNF quality testing and daily entry management.
- Smart delivery route optimization & live pickup/drop tracking.
- Customer subscription, ordering & QR-based milk traceability.
- AI-based regional demand forecasting.
- Real-time alerts for spoilage risk and delivery delays.
- Admin control panel for platform-wide oversight.
- Government / Cooperative portal for veterinary and compliance data.

Non-Functional Requirements:

- Real-time synchronization across all apps.
- Offline-first design for low-connectivity areas.
- Role-based, secure and permissioned access.
- Data security, encryption and privacy.
- Scalability across dairy cooperative networks.
- Responsive, user-friendly interface.

User Activities

USERS

- Admin
- Dairy Farmer
- Milk Collection Center Staff
- Delivery Partner
- Customer

- Customer Support Staff
- Dairy Cooperatives / Government Departments

Admin:

- Oversee platform operations through the Admin Control Panel.
- Manage users, roles and permissions across all stakeholder apps.
- Monitor system-wide collection, quality and delivery data.
- Coordinate with the Government / Cooperative Portal for compliance data.

Dairy Farmer:

- Log daily milk collection through the Farmer App.
- View the milk quality grade for each collection.
- Receive instant, UPI-based payments.
- Track income and payment history digitally.

Milk Collection Center Staff:

- Record Fat% / SNF quality tests at the collection point.
- Manage daily collection entries via the Collection Center Dashboard.
- Flag quality issues for review.

Delivery Partner:

- Access optimized delivery routes through the Delivery Partner App.
- Track live pickup and drop status.
- Receive real-time alerts for spoilage risk and delays.

Customer:

- Subscribe to and order milk through the Customer Marketplace App.
- Trace milk from farm to doorstep using QR-based tracking.
- Raise complaints or queries through customer support.

Customer Support Staff:

- Handle customer complaints and queries.
- Coordinate with delivery partners and collection centers to resolve issues.

Dairy Cooperatives / Government Departments:

- Access the Government / Cooperative Portal for veterinary and compliance data.
- Support pilot rollouts and cooperative-network scaling.

Ethical Measures

User Ethics: Accurate quality and payment reporting, respect for farmer and customer privacy, prevention of data misuse.

Business Ethics: Transparent digital payments and equal, fair treatment of all farmers and customers.

Regulatory Ethics: Compliance with dairy, food-safety and veterinary regulations via the Government / Cooperative Portal.

Data & Privacy: Role-based access, encrypted storage, and secure login for every stakeholder.

AI & Technology: AI-based demand forecasting with unbiased, secure data handling.
Development Ethics: Reliable, well-tested software built with an offline-first design and documented development practices.

Impact & Benefits

+25% Farmer income growth via reduced middlemen margins	**-30%** Milk spoilage reduction through better tracking	**3x** Faster farmer payments — hours instead of weeks	**-20%** Delivery cost cut through smart route optimization

Economic: Boosts rural income and creates a data-driven, efficient supply chain.
Social: Empowers small dairy farmers with fair pricing and financial inclusion.
Environmental: Reduced spoilage lowers wastage and the carbon footprint of transport.

Feasibility & Viability — Strengths and Challenges

Strengths:

- Direct farmer-to-consumer channel increases farmer income.
- Digital quality records (Fat%, SNF) build consumer trust.
- Familiar tech stack (React, Node.js) speeds development.
- UPI integration leverages India's mature digital payment rails.
- Scales via existing dairy cooperative networks.

Challenges & Mitigation:

- Farmer digital literacy → simple UI & local training.
- Rural network gaps → offline-first collection app.
- Trust & adoption → pilot with government cooperatives first.
- Data privacy → role-based access & encrypted storage.
- Cold-chain cost → phased rollout at high-volume centers.

Research & References

- Department of Animal Husbandry & Dairying, Ministry of Fisheries, Government of India — Annual Report.
- National Dairy Development Board (NDDB) — dairy value chain & cooperative data.
- FAO Dairy Development Reports — cold-chain loss & food-safety guidelines.
- AmritaMaa Software Requirement Specification (SRS) Workbook — project documentation.

Data Science Activities

AmritaMaa treats the daily stream of collection, quality and delivery data as a working asset — the sections below outline the core data science activities planned for the platform.

Predictive & Analytical Use Cases

- **Milk Quality Prediction —** regression models on Fat% / SNF readings to flag off-grade milk before it reaches the customer.
- **Demand Forecasting —** time-series models on order history to forecast regional demand and reduce overstock or shortage.
- **Spoilage Risk Alerts —** classification models on collection-to-delivery time and storage conditions to trigger real-time spoilage warnings.
- **Delivery Route Optimization —** route and clustering algorithms applied to pickup/drop data to cut delivery time and cost.
- **Farmer Income & Production Analytics —** state-wise and farmer-wise trend analysis to track income growth and production patterns.
- **Customer Insights —** analysis of subscription, ordering and complaint data to surface satisfaction trends and churn risk.

Data Pipeline

- Daily collection, quality-test and delivery entries flow from the Farmer, Collection Center and Delivery Partner apps into a central cloud database.
- Cleaned and structured data feeds the forecasting, quality-prediction and route-optimization models.
- Model outputs are surfaced back through the Admin Control Panel and stakeholder dashboards for day-to-day decisions.

Tools & Techniques

- Python (pandas, scikit-learn) for data cleaning, regression and classification models.
- Time-series forecasting for regional demand and seasonal production trends.
- Cloud-based analytics and dashboards (AWS / Firebase) for real-time reporting across stakeholders.

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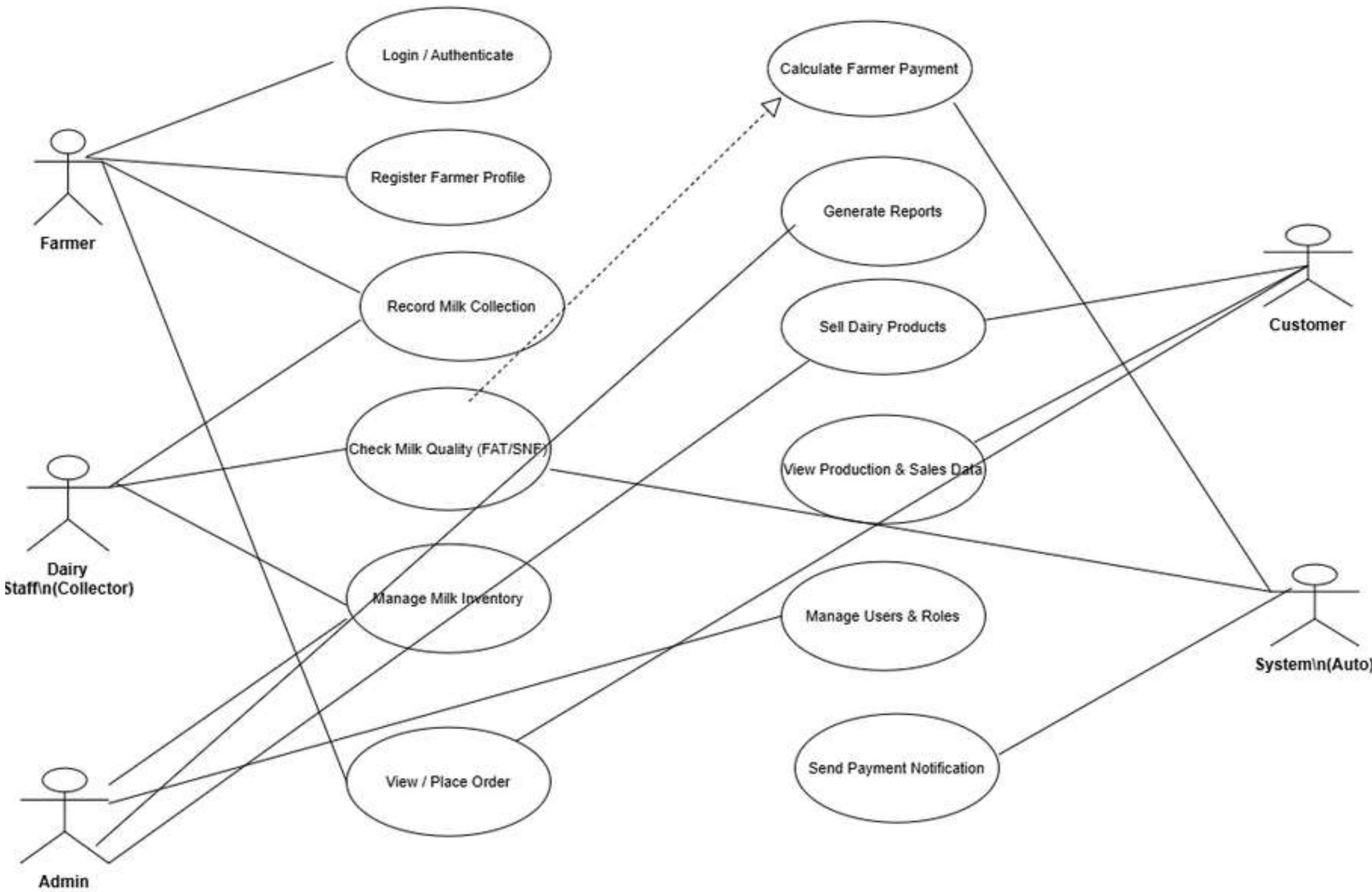
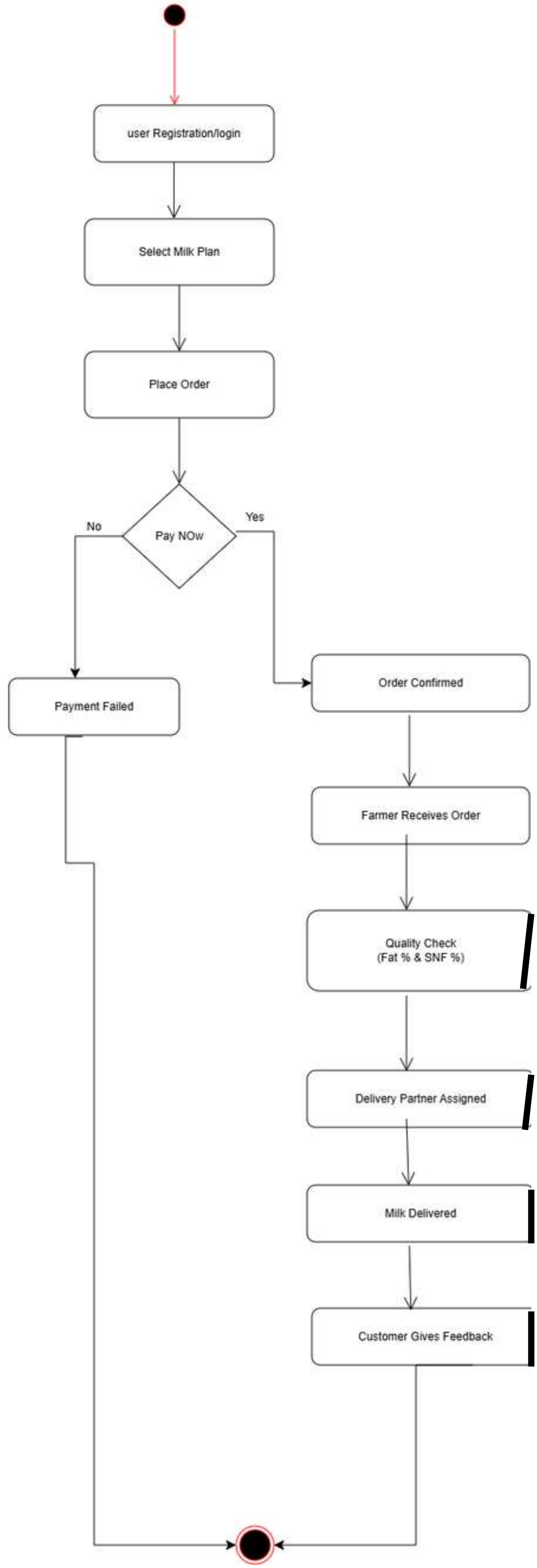
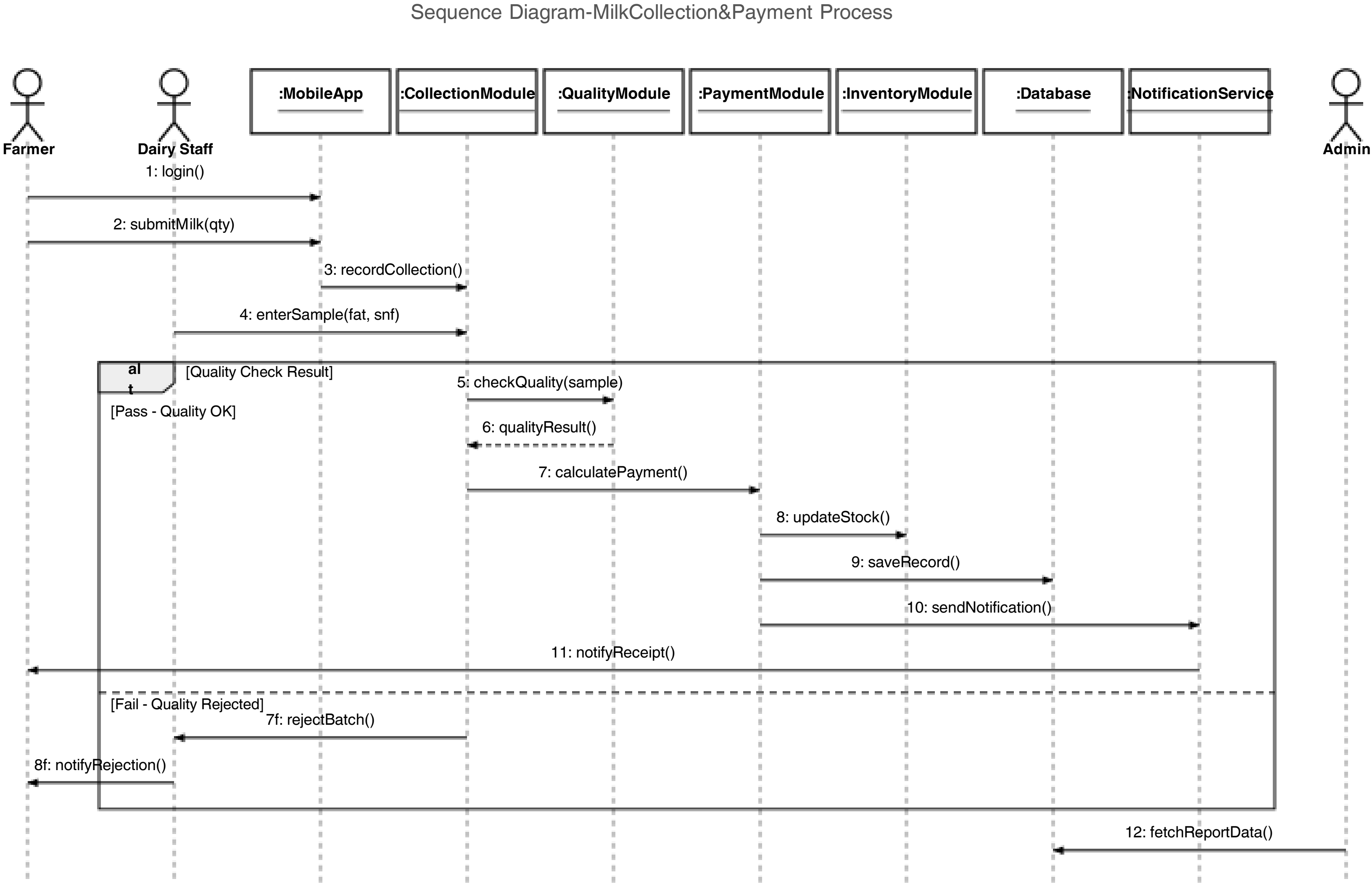
Use Case Diagram - Milk Production App




user Registration/login
Select Milk Plan
Place Order
Pay NOw
No
Yes
Payment Failed
Order Confirmed
Farmer Receives Order
Quality Check
(Fat % & SNF %)
Delivery Partner Assigned
Milk Delivered
Customer Gives Feedback



Legend: Solid arrow = call/message | Dashed arrow = return value | alt frame = conditional branch (Pass/Fail)

Amrit Maa - Milk Production Management App | Sequence Diagram

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CollaborationDiagram-MilkCollection&Payment Process